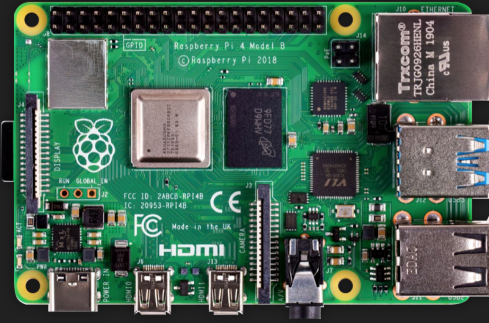


LCEVC and Raspberry Pi into the Metaverse!



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What's a Raspberry Pi?

- Extremely cheap and easy to obtain (\$35 worldwide)
- H.264 hardware encoder
- ARM64 compute
- Great supply chain
- Largely open source
- CSI camera input
- RAW Bayer from sensor into FFmpeg



What's LCEVC?

Low Complexity Enhancement Video Coding

ISO/IEC video coding standard developed by the Moving Picture Experts Group (MPEG)

V-Nova - (commercially available) codec implementations

Splits encoding into baseline with an “enhancement”

Enhancement is a delta between intended resolution and the baseline - rather than multiple full frame encodings

Why LCEVC?

(when compared to multiple separate encodings...)

Lower bitrate (sending delta can be 40% saving on bandwidth and so less overall costs to network operators and for CDNs quicker cache propagation and storage)

Multiple baseline codecs (H.264, HEVC, AV1...)

Lower CPU/GPU overall (potentially better battery life)

Faster to encode, can use older hardware

Easy(ish) to decode needs some compute

LCEVC Encoding with FFmpeg (*V-Nova pending PR*)

Baseline uses existing hardware encoder (typically $\frac{1}{4}$ res)

Enhancement layer(s) encode via CPU or if device has it,

NVIDIA CUDA accelerated

Multi-platform (arm64/x86_64), nvenc, x264...

FFmpeg example (it's not painful)

```
ffmpeg -y -i <input> c:v lcevc_h264 -base_encoder x264  
-eil_params 'preset=...' LCEVC.mp4
```

(Videon EdgeCaster also supports into 'video edge platform')

LCEVC Decoding (with Wasm/MSE/Canvas)

Shaka Player mainline open source on GitHub today!
(recently added PR - kudos Joey/Vinod and team)

Optimised WebAssembly performs all calculations
HTML5 canvas renders

Easy to integrate on other browser players and dash.js

(THEO also have commercial integration for this too)

LCEVC Metaverse

Works on Meta Quest too!

ExoPlayer edits due to be
open source soon(ish)

<https://github.com/v-novaltd>

Volumetric (cloud point data)
demo/development



What's next, where you might be able to help?

Raspberry Pi needs some love for H.264 live at profile/level
OpenMAX/libcamera implementation **could** handle (demo
performance is CPU only, so remains low res to keep up)

Hardware DRM has some challenges in the decoder due to
protected pipelines/trusted execution (resolution coming via W3C)

Watermarking has some challenges where player reconstructed
video blits out the watermark (open work with Nagra/Synamedia)

Potentially side band in-the-clear enhancement files (for 8K)